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// Circular Queue Implementation Using Array for Parcel Handling System

#include<stdio.h>
#define size 5
void main(){
    int arr[size];
    int op;
    int front = -1, rear = -1;
    do{
        printf("1. Add\n2. Dispatch\n3. Display\n4. Exit\n");
        printf("Enter a option: ");
        scanf("%d", &op);
        switch(op){
            case 1:
            if(front == (rear + 1)%size){
                printf("Parcel queue is Full\n\n");
                break;
            }else{
                if(front == -1){
                    front++;
                }
                rear = (rear+1)%size;
                printf("Enter parcel no. to add: ");
                scanf("%d", &arr[rear]);
                printf("\n");
            }
            break;

            case 2:
            if(front == -1){
                printf("Parcel queue is Empty\n\n");
                break;
            }else{
                printf("Parcel %d is dispatched\n\n", arr[front]);
                if(front == rear){
                    front = -1;
                    rear = -1;
                }
                front = (front+1)%size;
            }
            break;
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            case 3:
            if(front == -1){
                printf("Parcel queue is empty\n\n");
                break;
            }else{
                int i = front;
                printf("Parcel no. left in queue are: ");
                do{
                    printf("%d ", arr[i]);
                    i = (i+1)%size;
                }while(i != (rear+1)%size);
                printf("\n\n");
            }
            break;

            case 4:
            printf("EXITED SUCCESSFULLY");
            break;

        }
    }while(op <= 3);
}

//Output
1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 1
Enter parcel no. to add: 1

1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 1
Enter parcel no. to add: 2

1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 1
Enter parcel no. to add: 3
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1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 1
Enter parcel no. to add: 4

1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 1
Enter parcel no. to add: 5

1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 1
Parcel queue is Full

1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 2
Parcel 1 is dispatched

1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 3
Parcel no. left in queue are: 2 3 4 5

1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 4
EXITED SUCCESSFULLY